

Ehsan Ghoreishi

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SUMMARY

Ph.D. candidate in Computer Engineering (expected Feb. 2027) with 4+ years of 5G NR research experience in real-time wireless scheduling, physical-layer modeling, and AI/ML algorithms for 5G NR, O-RAN, and LEO satellite networks, using Python, C/C++, CUDA, and MATLAB.

SKILLS

Wireless: 5G NR, LTE, 3GPP, O-RAN, OFDM/OFDMA, MIMO, Beamforming, NTN/LEO, BER/BLER, Interference Analysis

Programming: Python, C/C++, CUDA, MATLAB, Simulink, Wireless System Simulation

AI/ML: Deep Reinforcement Learning, LSTMs, CNNs, PyTorch, TensorFlow, TF-Agents

Systems: Real-Time Algorithm Design, GPU Acceleration, Parallel Processing, Performance Evaluation, Debugging

EXPERIENCE

Graduate Research Assistant – 5G Real-Time Scheduling & O-RAN

2022–present

Complex Network & Security Research Lab, Virginia Tech

- Designed real-time URLLC/eMBB multiplexing algorithms for 5G NR and validated them under strict latency, reliability, and resource constraints; work led to a U.S. patent application and INFOCOM/IEEE TMLCN papers.
- Implemented GPU-accelerated decision making in C/C++ and CUDA, reaching $\sim 125 \mu\text{s}$ execution latency for time-critical applications.
- Built deep-RL and LSTM-based schedulers and integrated them into an O-RAN 5G architecture; evaluated system-level performance across diverse traffic and channel conditions.
- Built Python/MATLAB frameworks to simulate and analyze 5G behavior across traffic, channel, and scheduling scenarios.

Graduate Research Assistant – LEO Satellite Physical Layer & Beam Control

2025–present

Complex Network & Security Research Lab, Virginia Tech

- Developed system-level PHY and link-budget models for LEO satellite networks, including antenna gain, propagation loss, SNR, spectrum reuse, co-channel interference, beam allocation, and mobility.
- Designed handover-aware satellite and beam-selection algorithms; analyzed throughput, connectivity, and handover performance under mobility and capacity constraints.

Research Assistant – ML for Wireless Signal Classification

2020–2021

Signal Processing Research Center, Isfahan University of Technology

- Built signal-processing and neural-network pipelines on complex baseband signals for modulation classification; analyzed failure cases across signal conditions.

EDUCATION & AWARDS

Ph.D., Computer Engineering, Virginia Tech

Expected Feb. 2027

M.Sc., Computer Engineering, Virginia Tech

2024

B.Sc., Electrical Engineering, Isfahan University of Technology

2021

Award: CCI SWVA Cyber Innovation Scholar, Commonwealth Cyber Initiative

2025

PATENT & SELECTED PUBLICATIONS

- **U.S. Patent Application** (pending): “Multiplexing Ultra Low Latency Data with Other Data over the Same Radio Frame,” 2026.
- **E. Ghoreishi et al.**, “RaP: Learning-based Joint Reservation and Puncturing for Efficient URLLC/eMBB Multiplexing,” *IEEE INFOCOM*, 2026.
- **E. Ghoreishi et al.**, “Cyrus+: A Real-Time DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN,” *IEEE TMLCN*, 2025.
- **E. Ghoreishi et al.**, “Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN,” *IEEE ICCCN*, 2024 (Invited).
- 3 papers under review (INFOCOM 2027, ICC 2027); full list on Google Scholar.